```
clc;
clear;
close all;

% Required specifications
K = 2.5;
Ts = 8;

% Calculate time constant
tau = Ts/4;

% Create transfer function
G = tf(K, [tau 1]);

% Display results
disp('Transfer Function:');
G

disp('Time Constant:');
tau

disp('System Pole:');
pole(G)

disp('DC Gain / Final Value:');
dcgain(G)

disp('Step Response Information:');
stepinfo(G)

% Plot unit-step response
figure;
step(G);
grid on;
title('First-Order System Response');
xlabel('Time (seconds)');
ylabel('Output');
```

```
Transfer Function:

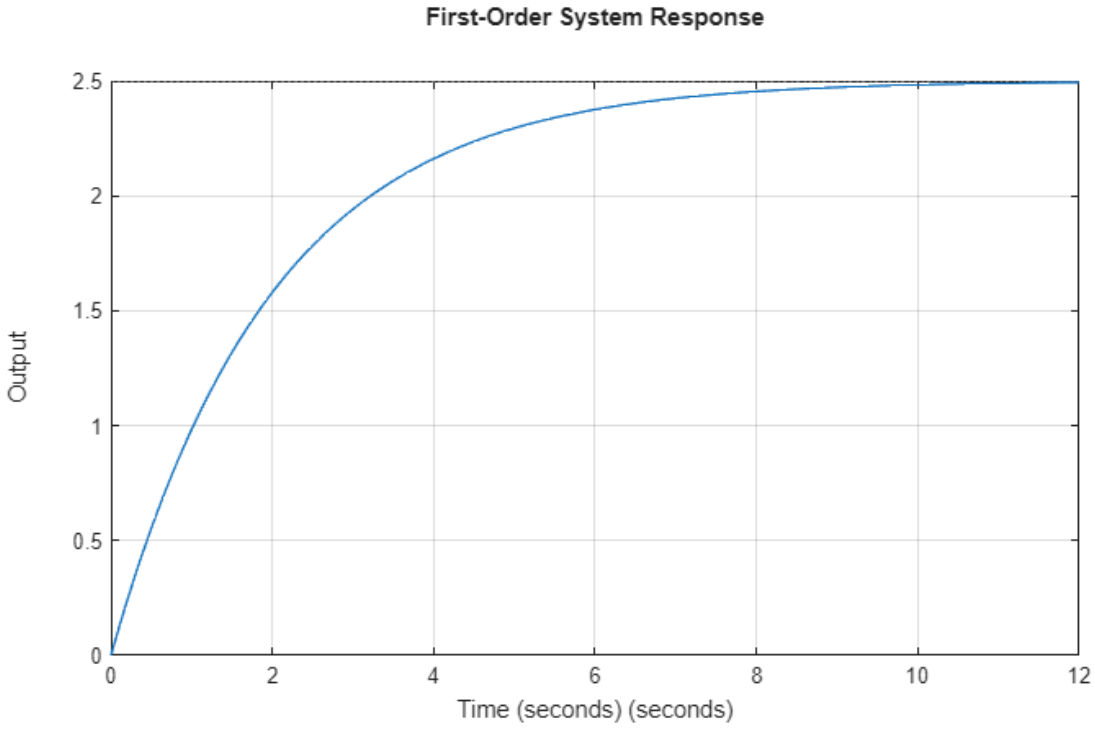
G =

    2.5
  -------
  2 s + 1

Continuous-time transfer function.
Time Constant:
tau =
     2
System Pole:
ans =
   -0.5000
```

```
DC Gain / Final Value:
ans =
    2.5000
Step Response Information:
ans =
  struct with fields:

         RiseTime: 4.3940
    TransientTime: 7.8241
     SettlingTime: 7.8241
      SettlingMin: 2.2613
      SettlingMax: 2.4983
        Overshoot: 0
       Undershoot: 0
             Peak: 2.4983
         PeakTime: 14.6444
```



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